Dhamija, Gaurav; Gupta, Sagnik Kumar; Ojha, Manini

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The Protective Power of Connectivity: Internet Exposure and Intimate Partner Violence

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The Protective Power of Connectivity: Internet Exposure and Intimate Partner Violence

Gaurav Dhamija * Sagnik Kumar Gupta † Manini Ojha ‡

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Abstract

We study the effect of women's exposure to the internet on the incidence of intimate partner violence (IPV) utilizing data from a large-scale nationally representative data for India. Utilizing an instrumental variable approach, we exploit the exogenous variation arising from cellular tower density at the district level as an instrument for internet exposure. We find women who have access to the internet are at a significantly lower risk of exposure to physical, sexual, emotional and any IPV by 17, 5, 9 and 18 percentage points.We provide further indicative evidence that women's attitudes justifying violence works as a channel of our estimated causal impacts. Our results are robust to a host of sensitivity checks, additional controls, alternative definitions and falsification analysis. We document important heterogeneity in the effects across social groups, wealth index, area of residence, and women's education levels. The effects appear to be particularly important for women from wealthier households and those who are more educated.

JEL Classifications: I31, J12, J16, O12, O33

Keywords: Internet Exposure, Tower Density, Intimate Partner Violence, Physical Violence, Sexual Violence, Emotional Violence, Instrumental Variable, NFHS-5, India

Ordering of authors: All authors contributed equally; author order was determined randomly.

*Indian Institute of Technology Hyderabad. Email: gauravdhamija@la.iith.ac.in

†Indian Institute of Technology Hyderabad. Email: la24resch11001@la.iith.ac.in

‡Jindal School of Governement and Public Policy, O. P. Jindal Global University, CDES Non-Resident Fellow; GLO Fellow. Email: mojha@jgu.edu.in

1 Introduction

Violence against women (VAW), in general, and intimate partner violence (IPV), in particular, remains one of the most pervasive human rights violations globally, with grave consequences for women's health, autonomy, and well-being (World Health Organization, 2019). Despite growing public attention, a significant share of women continue to feel unsafe within their own homes, often the very spaces assumed to provide security (World Health Organization, 2021). IPV stems from and reinforces deeply entrenched patriarchal norms, male dominance, unequal bargaining power, and cultural acceptance of violence as a means of control in intimate relationships. The consequences of IPV are far-reaching. Beyond the immediate physical and psychological harm (Dicola and Spaar, 2016), IPV affects women's reproductive health, fertility decisions, educational attainment, labour force participation, mobility, and overall quality of life (Ackerson and Subramanian, 2008; Boy and Salihu, 2004; Yount et al., 2011; Babbar and Ojha, 2024; Campbell, 2002; Durevall and Lindskog, 2015; Ellsberg et al., 2008). It reduces women's capacity to participate fully in economic, social, and political spheres, and undermines broader goals of gender equity and human development. The United Nations' Sustainable Development Goals (SDGs), particularly Goal 5 on gender equality and Goal 16 on peace, justice, and strong institutions highlight the urgency of addressing violence against women not just as a private matter, but as a central development and policy concern.

While much of the existing literature has focused on individual-level predictors of IPV, or the role of education, employment, or legal reforms in mitigating violence, less is known about how digital connectivity might shift the dynamics of power, awareness, and resistance within households. With India undergoing one of the fastest digital transformations in the world, this raises a timely and important question - *can internet access reduce women's vulnerability to intimate partner violence?* The internet has the potential to function much like education in transforming women's lives. It can enhance awareness, access to information, connection to peer networks and support groups, and exposure to alternative value systems and role models. It may also enable women to challenge internalised patriarchal norms, recognise abuse, seek help, or even strengthen their bargaining power within relationships. However, it could also trigger backlash in contexts where female autonomy is contested, making the net effect an empirical question.

In this paper, we examine whether and to what extent women's exposure to the internet influences their likelihood of experiencing domestic abuse, using nationally representative data from India. We focus on three distinct forms of IPV, captured in the National Family Health Survey 2019-21 data, viz., physical, sexual, and emotional violence, as well as an aggregated indicator capturing exposure to any form of IPV. We aim to contribute to a growing but still nascent literature on the intersection of digital access, gender empowerment, and household power dynamics. Our study provides causal evidence by exploiting plausibly exogenous variation in district-level internet infrastructure in India, instrumented using mobile tower density, and sheds light on a critical yet under-explored dimension of the digital divide, that is, the impact of internet access on women's safety and autonomy within their homes.

This paper contributes to several interrelated strands of literature. First, it advances the growing body of work at the intersection of the economics of household bargaining and IPV. Theories of intra-household bargaining (Tauchen et al., 1991; Farmer and Tiefenthaler, 1996) emphasise that women's access to economic opportunities and control over resources shape their bargaining position within households, thereby influencing their vulnerability of IPV. Building on this framework, empirical studies have identified a variety of mechanisms through which shifts in women's bargaining power affect IPV outcomes. These channels include women's education (Erten and Keskin, 2018, 2020; Weitzman, 2018), employment opportunities (Erten and Keskin, 2021; Dhanaraj and Mahambare, 2022), gender wage gap (Aizer, 2010), political representation (Lnu et al., 2022), legal awareness (Erten and Keskin, 2022), and self-help groups (Sato et al., 2022). Institutional and demographic shifts such as reforms in divorce laws (Stevenson and Wolfers, 2006), evolving gender ratios (Amaral and Bhalotra, 2017), participation in protests (Stern and Erten, 2024), variations in marriage type (Roychowdhury and Dhamija, 2022), and age at marriage (Roychowdhury and Dhamija, 2021) have also been linked to women's exposure to IPV. More recent research has incorporated attitudinal and behavioural dimensions, such as norms justifying IPV (Mookerjee et al., 2021) and contraceptive decision-making (Babbar and Ojha, 2024), thereby broadening our understanding of how shifts in agency and norms influence IPV.

Second, this study contributes to the emerging literature on the transformative potential of the internet on a wide range of socioeconomic outcomes. Prior research has shown that internet access can delay age at marriage (Zhang et al., 2023; Murray, 2020), improve maternal and reproductive health outcomes (Krishnatri and Vellakkal, 2024), enhance labour market participation and earnings, particularly for women (Bahia et al., 2024; Jain and Chatterjee, 2024; Winkler, 2017). Internet exposure has also been linked to greater dietary diversity and nutritional knowledge (Vatsa et al., 2023; Deng et al., 2024), and higher civic participation (Donati, 2023). Yet, these benefits remain unevenly distributed. Women's digital participation is constrained not only by affordability, literacy, and skills but also by deeply entrenched gender norms that frame digital technology as inappropriate or unsafe for women (Dhawan, 2023). As noted by Kumawat and Garg (2025) (p. 976), "technology in women's hands is seen as an object of distrust and must be monitored to comply with social and cultural norms". Such restrictions reinforce existing inequalities, creating fear and self-censorship that limit digital engagement. Against this backdrop, women's access to the internet holds transformative potential to challenge internalised patriarchy and disrupt regressive social norms.

Third, our paper contributes to the literature examining how structural changes in the economy shape women's exposure to IPV. Previous research has analysed the role of trade liberalisation (Chong and Velásquez, 2024), urbanisation (Dhamija et al., 2025), electrification (Bhukta et al., 2024), and financial inclusion (Bhukta et al., 2025; Shreemoyee et al., 2025a). We extend this line of inquiry by conceptualising women's access to the internet as an emerging dimension of structural and informational change. By linking digital connectivity to women's attitudes toward IPV, we demonstrate how technological expansion can alter the risk of IPV.

We find that internet exposure significantly reduces women's risk of experiencing IPV. In our preferred instrumental variable (IV) specification, internet access lowers the likelihood of physical violence by 17 percentage points (pp), sexual violence by 5 pp, emotional violence by 9 pp, and any form of IPV by 18 pp. These effects are robust across a range of specifications, trimming thresholds, and additional controls, including husband's characteristics and childhood exposure to inter-parental violence. We also find suggestive evidence that the mechanism operates through changes in women's attitudes justifying IPV, with internet access reducing the likelihood of accepting such norms by 21 pp, and these attitudinal changes being strongly associated with lower IPV risk. We also document interesting heterogeneity across social groups, wealth, area of residence and women's education. We even note considerable heterogeneity across regions of India. Notably, the effects are more pronounced among wealthier and more educated women, while estimates for poorer and less educated women are less precise, driven in part by larger standard errors rather than negligible effect sizes.

This pattern suggests that while better-resourced women may currently be better positioned to convert digital access into safety and empowerment, the potential benefits for more marginalized women remain substantial but under-realized. From a policy perspective, this stresses the importance of not simply scaling up digital infrastructure, but intentionally designing digital inclusion policies such that target women at the margins, those with lower education, limited mobility, and weaker bargaining power within households.

2 Background

India's digital transformation has followed a long, and uneven trajectory. It has been shaped by the changing nature of policy priorities, market forces, and innovations in infrastructure. In the early decades, access to the internet was limited to research institutions and government agencies via networks such as Education and Research Network (ERNET) and National Informatics Centre Network (NICNET). Commercial availability only began in the mid-1990s, when the state-owned Videsh Sanchar Nigam Limited (VSNL) launched public internet services (Vinayak, 2020). This was initially restricted to metropolitan cities and elite users, despite the entry of private players in the late 1990s. There were still constraints such as high data costs, limited bandwidth, and sparse infrastructure. By 2008, only around 11 million internet subscribers existed nationwide which is modest given the size of the country (Mangla and Singh, 2021).

In 2010, more efforts to modernise connectivity gathered momentum, particularly with the launch of the Digital India initiative in 2015 (TOI Business Desk, 2025). This aimed to improve digital infrastructure and expand rural access. Finally, the entry of Reliance Jio in 2016 disrupted the market by offering low-cost 4G data and free voice services (Krishnatri and Vellakkal, 2024). This triggered a massive decline in data prices fell by over 90% within a year and brought mobile internet within reach for millions of new users. Between 2016 and 2021, the number of wireless internet subscribers more than doubled, with over 96% of broadband users accessing the internet via wireless devices by late 2021 (Telecom Regulatory Authority of India, 2022). The dominance of mobile-based access, especially in regions without wired connections, placed renewed importance on the geographic distribution of telecom infrastructure, particularly mobile towers.

In this context, the density of mobile towers emerged as a crucial determinant of internet availability. Initially, telecom operators built and maintained their own towers, limiting their coverage to only profitable urban centres. This began to shift with the adoption of the Towerco model, under which independent firms built and operated towers that could be leased to multiple service providers (Azam et al., 2024). This model significantly reduced deployment costs and enabled operators to scale more rapidly in previously underserved rural areas. By 2020, over 84% of India's towers were managed by such independent firms, and the total number of base stations had increased dramatically from under 200,000 in 2008 to over 22 million by 2020 (Houngbonon et al., 2021; Azam et al., 2024). Reliance Jio's rapid expansion illustrates this infrastructural pivot. By 2020, it had rolled out its own vast network of 4G towers, extending mobile broadband access to remote villages and districts with limited electrification. As a result, regional disparities in internet access became closely tied to district-level tower density, which now serves as a credible proxy for measuring the intensity of digital connectivity across geographies.

3 Data

For our analysis, we utilize the most recent wave of the National Family Health Survey (International Institute for Population Sciences (IIPS) and ICF, 2021) conducted in India during 2019-21. The NFHS is a nationally representative, cross-sectional demographic health survey, forming part of the global Demographic and Health Survey (DHS) program.[1] The survey is implemented by the International Institute for Population Sciences (IIPS), Mumbai, under the supervision of the Ministry of Health and Family Welfare (MoHFW), Government of India. The sample is drawn using stratified random sampling.[2] A women's questionnaire was administered to all eligible women aged 15–49 residing in 636,669 sampled households. It collected detailed information on various aspects of gender and household dynamics, including demographic characteristics, fertility and contraception, nutrition, marriage, sexual behaviour, employment, experiences of domestic violence, mobility, and autonomy. However, certain topics, such as sexual behaviour, attitudes and experiences of domestic violence, were included only in a randomly selected subsample of about 15 percent of households through a state-specific module. Following the World Health Organization's ethical guidelines for collecting data on domestic violence, only one eligible woman per household was randomly selected to participate in the domestic violence module. The module was administered to 72,320 women, with strict procedures to ensure respondents' privacy during the interview process.

Our outcome variables capture women's exposure to IPV, categorized into three distinct forms: physical, sexual, and emotional. A detailed classification of the specific acts included under each type is provided in Table A1. Each category takes the value 1 if the respondent reports experiencing at least one of the underlying acts of that form of violence in the past twelve months, and 0 otherwise. Following prior literature (Roychowdhury and Dhamija, 2022; Shreemoyee et al., 2025a), we also construct a composite measure, $AnyIPV$, which equals 1 if a woman reports experiencing at least one of the three types of IPV, and 0 otherwise. Thus, our four primary outcome variables are $PhysicalIPV$, $SexualIPV$, $EmotionalIPV$, and $AnyIPV$.

The primary variable of interest in our analysis is women's access to the internet, referred to as $InternetExposure$. This is a binary variable coded as 1 if the woman reports ever having

[1] The DHS surveys for all countries are available at https://dhsprogram.com/.

[2] See International Institute for Population Sciences (IIPS) and ICF (2021) for more details on the survey methodology.

used internet, and 0 otherwise. To mitigate concerns of omitted variable bias and isolate the causal effect of internet exposure on women's experiences of IPV, we include a rich set of control variables following the approach suggested by Oster (2019). These controls account for potential confounders at the individual, household, and district levels. Individual-level controls include the woman's age (in years), completed years of education, employed during the last twelve months (yes if working and no otherwise), current marital status (yes if currently married and no if widowed, divorced, or separated), exposure to mass media (yes if exposed to newspaper, radio or television and no otherwise), ownership of a house or agricultural land (yes if sole or joint ownership and no otherwise), and duration of residence in the current location (in years). At the household level, we control for gender of the household head (yes if male-headed and no otherwise), age of the head (in years), number of household members, religion (Hindu, Muslim, or other), social group (Scheduled Caste [SC], Scheduled Tribe [ST], Other Backward Class [OBC], other social group/none, or don't know/missing), wealth status (poor/non-poor), and type of residence (rural/urban).

We further include district-level socioeconomic and geographic controls to capture broader contextual variation that may influence both internet access and IPV exposure. We obtain district-level socioeconomic and geographic indicators from the Socioeconomic High-resolution Rural–Urban Geographic Platform for India (SHRUG), an openly accessible dataset that provides detailed information on multiple dimensions of development across more than 600,000 Indian villages and towns.[3] The NFHS-5 data are subsequently linked with SHRUG measures at the district level, yielding complete information for 692 districts.

The district-level socioeconomic indicators include the following: the sex ratio (females per male), share of SC and ST population, and the availability of key public infrastructure such as family welfare centres, maternity and child welfare centres, and educational institutions at different levels—primary, middle, and secondary schools, as well as medical, engineering, arts and science, and vocational training colleges. SHRUG has drawn these indicators from the District Census Abstract of the 2011 Census of India, which aggregates information from all villages and towns within each district.[4]

The geographical characteristics used as controls include measures of elevation, terrain ruggedness, nighttime light intensity, frequency of lightning events, and vegetation cover. Elevation data are sourced from the Shuttle Radar Topography Mission (SRTM) with a 1-arc-second resolution. Ruggedness captures the local variation in elevation between neighbouring surfaces

[3]The SHRUG database is publicly accessible at http://devdatalab.org/shrug. See Asher et al. (2021) for further details.

[4]The District Census Abstract provides the compiled number of public institutions across villages in each district.

(Riley et al., 1999; Nunn and Puga, 2012). Nighttime lights, widely used as a proxy for regional economic activity and urbanisation (Dhamija et al., 2025), are measured as the mean annual cloud-free radiance values derived from Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band (DNB) data, collected jointly by National Aeronautics and Space Administration (NASA) and National Oceanic and Atmospheric Administration (NOAA) (Elvidge et al., 2017; Mayala et al., 2018; Mayala and Donohue, 2022). Information on lightning frequency, the total number of lightning days between 1969-2019, is obtained from the India Meteorological Department's Annual Disaster Weather Report (1969–2019).[5] To minimise the influence of extreme values, all continuous geographical variables are expressed in their natural logarithmic form. Finally, vegetation continuous fields represent annual tree cover, proportion pixels classified as forested in a district, estimated using a machine learning model that integrates NASA MODIS imagery with higher-resolution satellite samples (Townshend et al., 2011). To mitigate potential simultaneity concerns, we use one-year lagged values of nighttime lights and vegetation continuous fields corresponding to the survey years 2019–21 for each district.

We utilize the individual recode dataset from NFHS-5, which provides one observation per eligible woman respondent. The domestic violence module of the survey includes information on 72,320 women aged 15–49 years. Our analytical sample is constructed in three stages. First, we drop 8,469 women who were not married. Second, we exclude 1,278 women who were listed as visitors in the household roster. Together, these exclusions yield a sample of 62,573 ever-married women who are usual residents of their respective households. Third, we restrict the dataset to women with complete information on all variables used in the analysis. After excluding observations with missing data on years of residence at the current location (45), age of the household head (10), and district-level socioeconomic characteristics (1,256), our final analytical sample consists of 61,262 women. A schematic overview of the sample construction is available in Figure A1.

Table 1 presents descriptive statistics for individual- and household-level variables. Within our analytical sample, 22% of women have reported experiences of physical IPV, 5% have reported sexual IPV, 11% have reported emotional IPV, and 25% have reported any form of IPV within the past twelve months. On average, women are 34.1 years old, have 6.4 years of completed education, and have resided in their current location for about 14.9 years. About 27% of women have used the internet, 95% are currently married, 73% have some exposure to mass media, and 52% report ownership of land or house. At the household level, 84% of households have male heads, with an average age of 46.2 years, and 4.9 members. Around,

[5] See https://imdpune.gov.in/hazardatlas/lightvulabout.html.

76% of households are Hindu, 19% belong to SC, 19% to ST, 38% to OBC, 18% to other social groups, 45% fall into the poorest or poor category of DHS wealth index[6], and three-quarters (76%) resides in rural areas.

Table 2 provides district-level averages for key covariates. As per the 2011 census data, districts on average had 1,629 primary schools, 743 middle schools, and 327 secondary schools, 2 medical colleges, 6 engineering colleges, 15 arts and science colleges, 14 vocational training institutes, 107 family welfare centres, and 64 maternity and child welfare centres per district. SC and ST together constituted roughly one-third (33%) of the district population, and the sex ratio stood at 95 women for every 100 men.

4 Empirical framework

4.1 Ordinary least squares Estimation

We begin by estimating the following empirical model to investigate the effect of women's access to the internet on their likelihood of experiencing IPV:

$$Y_i \quad = \quad \beta_0 + \beta_1 InternetExposure_i + \beta_2 \mathbb{X}_{ih} + \beta_3 \mathbb{G}_d + \beta_4 (\mathbb{S}_d \times t) + \theta_s + \epsilon_{ihdst} \tag{1}$$

In this specification, Y_i refers to the binary indicators of physical, sexual, emotional, and any form of IPV for woman i. $InternetExposure_i$ measures the woman i's reported use of the internet. The vector $\mathbb{X}_{ih}$ includes individual- and household-level controls. District-level geographic characteristics, $\mathbb{G}_d$, are included to adjust for ecological and infrastructural conditions that directly influence the feasibility of extending internet infrastructure. $\mathbb{S}_d \times t$ denotes district-level socioeconomic characteristics interacted with survey year t to absorb potential time-varying effects. The state fixed effects, θ_s, account for any systematic state-level time-invariant unobservables that may affect internet exposure and IPV attitudes. ϵ_{ihdst} is the idiosyncratic error term. All standard errors are clustered at the cluster level.

Our parameter of interest, β_1, captures the effect of women's internet access on their exposure to IPV. A causal interpretation of β_1 would require that there are no unobserved variables that are correlated with both our variable of interest and our outcome variable, conditional on our controls. The presence of endogeneity would render a simple OLS estimate of β_1 to be biased and inconsistent. For instance, women's unobserved ability is likely to affect women's access to internet (Sina et al., 2023) and might be correlated with their IPV exposure (Cherrier

[6]The DHS wealth index is a standardized measure of household economic status that divides households into five quantiles: poorest, poor, middle, richer, and richest.

et al., 2023). Consequently, the OLS estimates may be biased due to the omission of unobserved confounding factors, reverse causality, or on account of measurement error. To alleviate this concern, we adopt an IV approach to estimate the effect of a woman's access to the internet on her IPV exposure.

4.2 Instrumental variable estimation

We exploit spatial variation in digital infrastructure as an IV for women's access to the internet. Specifically, we use the district-level mobile tower density as a proxy for the intensity of mobile network coverage. The data are sourced from OpenCellID, a global, open-source database that compiles information on the cellular tower locations through a combination of crowd sourced and publicly available inputs.[7] Each record in the dataset provides the geographical coordinates of Base Transceiver Stations (BTS). As of mid-August 2020, the OpenCellID platform documented approximately 2.36 million BTS across India, a count that aligns closely with Telecom Regulatory Authority of India (TRAI) figures reporting 2.31 million BTS in March 2020 (Telecom Regulatory Authority of India, 2021).

We spatially link the tower location data to district boundaries defined by the DHS shapefile and aggregate them to create a district-level measure of tower concentration. The variable $TowerDensity$ represents the number of BTS within a district divided by its land area. In line with other geographical characteristics, we minimise the influence of extreme values by taking the natural logarithm of ($1 + TowerDensity$) before estimation. Henceforth, we refer to our instrument variable, $log(1 + TowerDensity)$, as $AdjustedTowerDensity$. In our analytical sample, districts on average contain 3,124 towers, corresponding to a mean density of 3.1 towers per square kilometre. As shown in Figure 1, there is substantial spatial heterogeneity in $TowerDensity$ across India. The distribution of $AdjustedTowerDensity$ is presented in Figure 2.

We estimate a two-stage IV model, which is specified as follows:

$$InternetExposure_i = \alpha_0 + \alpha_1 AdjustedTowerDensity_d + \alpha_2 \mathbb{X}_{ih} + \alpha_3 \mathbb{G}_d + \alpha_4(\mathbb{S}_d \times t) + \phi_s + \eta_{ihdst} \tag{2}$$

$$Y_i = \beta_0 + \beta_1 InternetExposure_i + \beta_2 \mathbb{X}_{ih} + \beta_3 \mathbb{G}_d + \beta_4(\mathbb{S}_d \times t) + \theta_s + \epsilon_{ihdst} \tag{3}$$

[7] See https://datacatalog.worldbank.org/search/dataset/0038043/Global-OpenCellID-cell-tower-map and https://www.opencellid.org/ for details.

Equations (2) and (3) represent the first-stage and structural specifications of our model. Women's internet exposure, $InternetExposure_i$, is instrumented using district-level adjusted tower density, $AdjustedTowerDensity_d$. The model adjusts for individual and household characteristics ($\mathbb{X}_{ih}$), district-level geographic factors ($\mathbb{G}_{ds}$), district socioeconomic characteristics from the 2011 Census interacted with survey year ($\mathbb{S}_{ds} \times t$), and state fixed effects (ϕ_s or θ_s). ϵ_{ihdst} is the idiosyncratic error term. The coefficient of interest, β_1 captures the effect of woman's internet access on her exposure to IPV.

Following (Babbar and Ojha, 2024; Gershenson et al., 2022; Chowdhury et al., 2018; Trinh, 2020; Addison and Teixeira, 2019), we estimate the model using the Conditional Mixed Process (CMP) framework of Roodman (2011). This framework is suitable for systems involving binary endogenous and outcome variables. The CMP estimator efficiently accounts for the limited nature of the first-stage dependent variable. Moreover, it is likely to offer greater precision relative to linear instrumental variable models (Roodman, 2011). As noted by Roodman (2011), CMP operates as a form of seemingly unrelated regression (SUR) and can be applied to a wide range of simultaneous equation systems. In our setting, CMP functions as a limited-information maximum likelihood (LIML) estimator, with the structural interpretation confined to the coefficients from the second stage. The general likelihood formulation for this estimator is detailed in Roodman (2011).

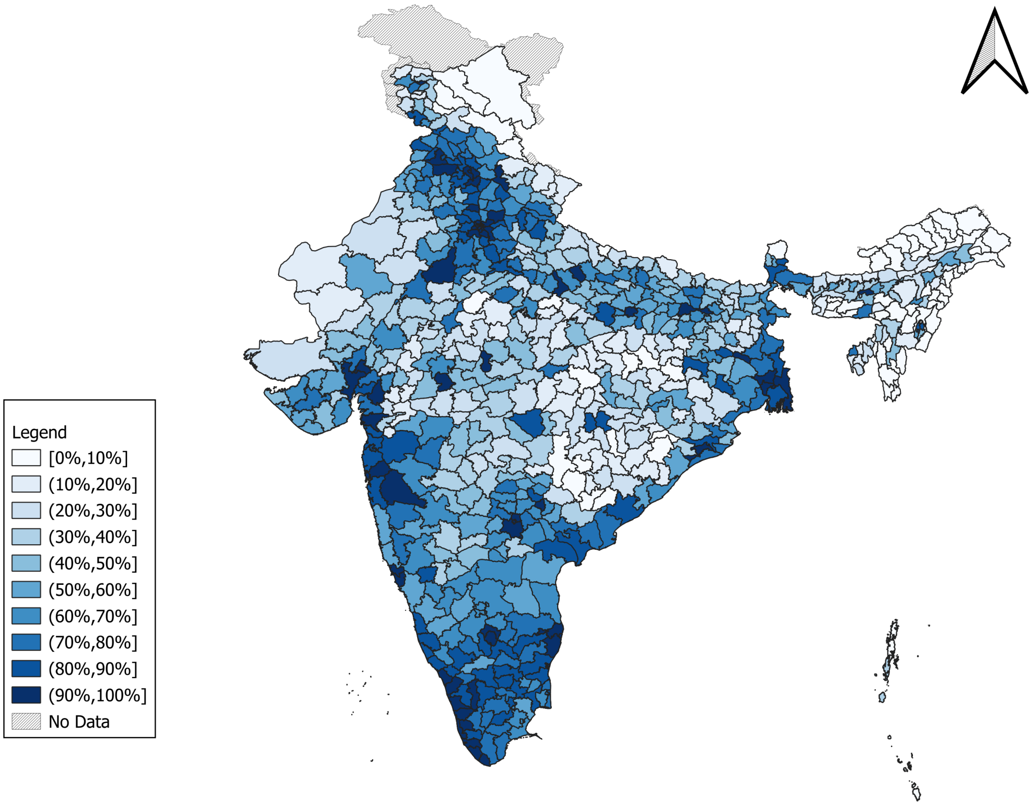


Figure 1: District-level map of Tower Density.
Notes: Plotted by authors using district coordinates from DHS as well as Survey of India and tower data is sourced from OpenCellID

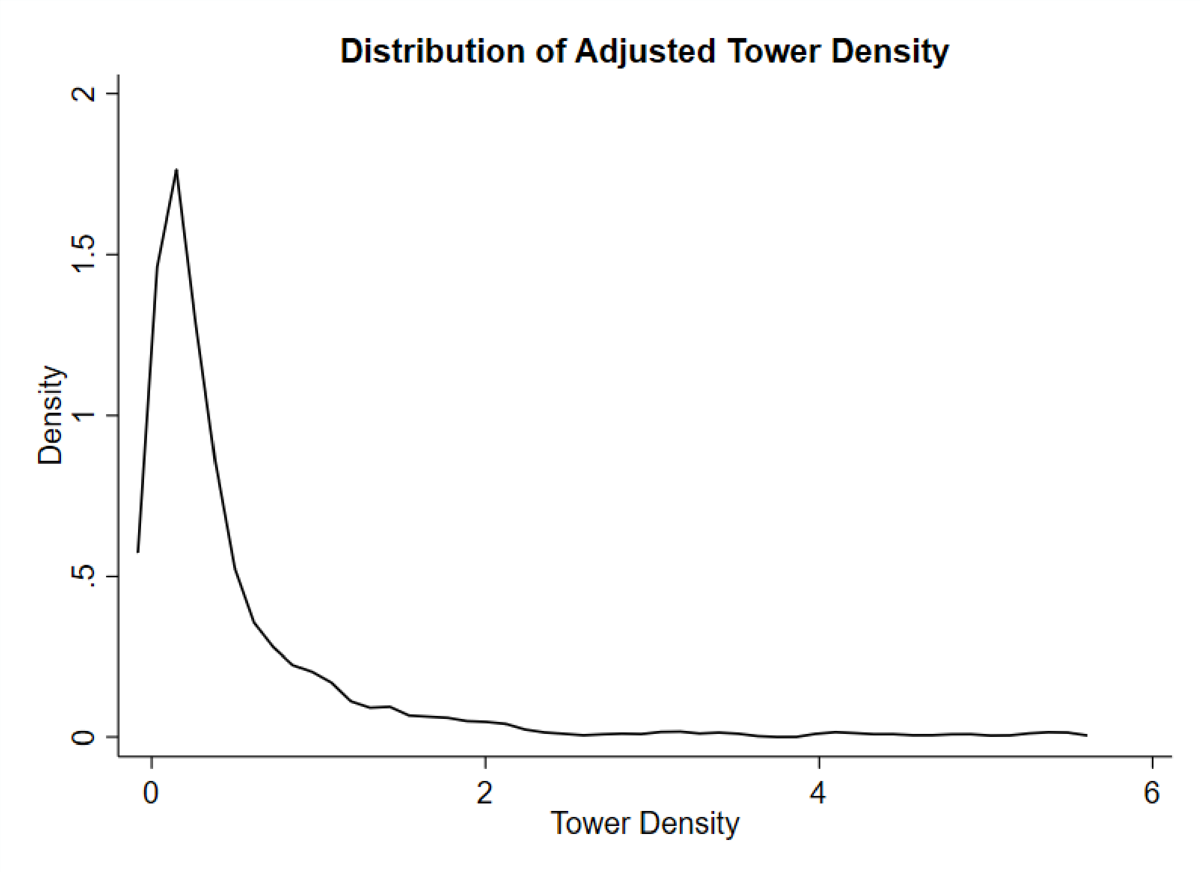


Figure 2: Kernel density plot of adjusted tower density.

4.3 Validity of the IV

We employ three approaches to assess the validity of adjusted tower density as an instrument for women's internet exposure. First, we examine the relationship between the two by plotting the district-level averages of women's internet use against adjusted tower density. Figure 3 illustrates a strong positive correlation, providing preliminary evidence that districts with greater tower density exhibit higher levels of women's internet exposure.

Second, Table 3 presents our first-stage estimates based on Equation (2). In Column (1), we begin by examining the relationship between women's internet access and district-level adjusted tower density without including any controls. We use the adjusted tower density to facilitate interpretation. The results show a strong and statistically significant association: a 1% increase in tower density is linked to a 11 percentage point rise in the probability that a woman has ever used the internet. As we add controls across columns (2) to (4), starting with individual and household characteristics, then geographic and socioeconomic characteristics at the district level, and finally, state fixed effects, we observe a reduction in the magnitude of the coefficient. However, the positive and statistically significant relationship between adjusted tower density and internet exposure remains robust throughout. Our preferred specification is shown in Column (4), which includes the full set of controls along with state fixed effects. Here, we find that a 1% increase in tower density is associated with an 8 percentage point increase in the likelihood of internet use among women. We also assess the strength of our instrument. The Kleibergen-Paap rk LM statistic (41.37) and the first-stage F-statistic (42.97) provide support for the relevance of our instrument. Across all specifications, the F-statistic comfortably exceeds the conventional threshold of 10, helping rule out concerns around weak identification.

Third, we argue that our instrument affects women's IPV exposure only through its impact on internet exposure, conditional on the covariates included in the IV specification. A potential concern is that the socioeconomic structure of districts may influence both tower deployment and women's exposure to IPV. To address this concern, we implement a residual-based diagnostic to visually assess whether adjusted tower density is associated with IPV exposure after accounting for internet exposure. Specifically, we regress women's exposure to $AnyIPV$ on internet exposure and save the residuals, and similarly obtain residuals from regressing adjusted tower density on internet exposure. Figure 4 plots these residuals and shows no discernible pattern. It suggests that, conditional on internet exposure, adjusted tower density is not systematically related to women's IPV exposure. Additionally, our empirical model includes a comprehensive set of controls for both geographic and socioeconomic characteristics, as well as state fixed effects, which together substantially reduce the risk of violation of exclusion restriction.

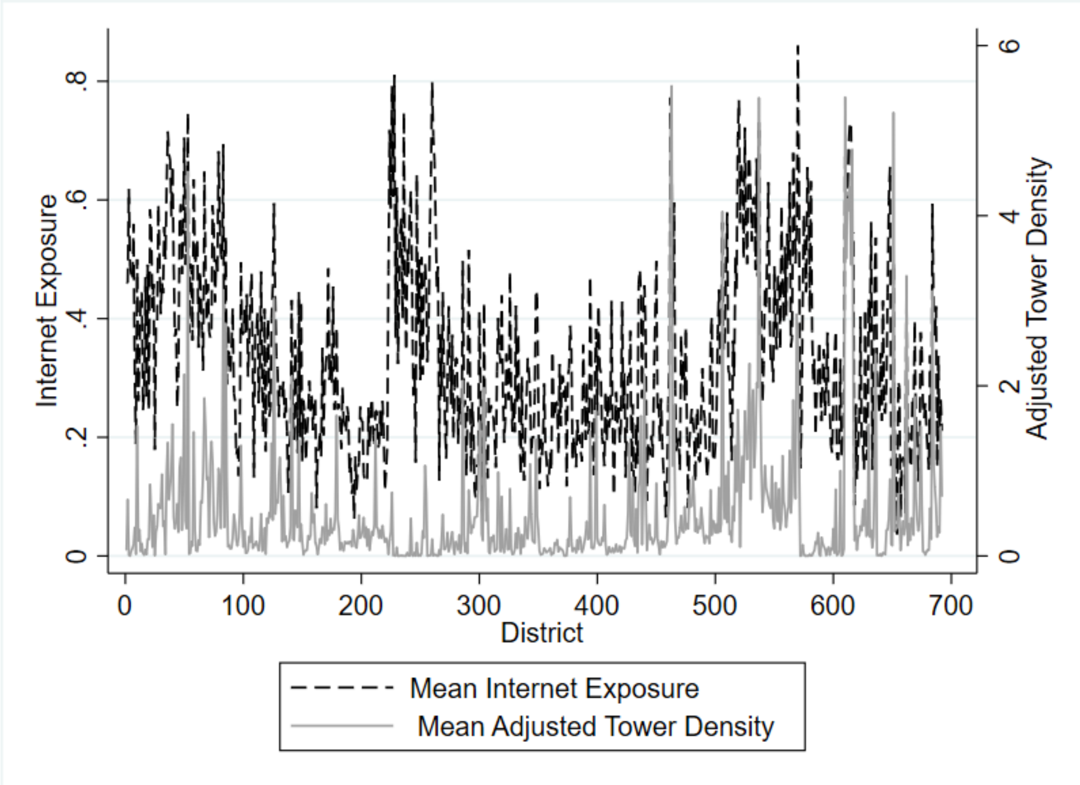


Figure 3: First stage correlation between average district level internet exposure and district level adjusted tower density

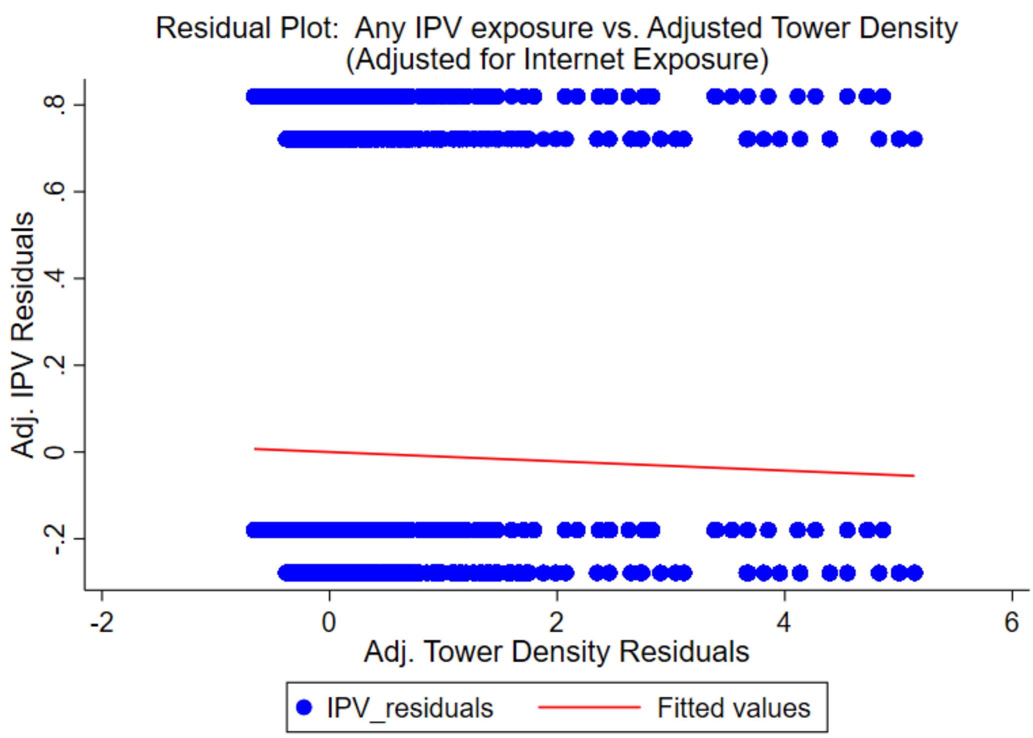


Figure 4: Residual plots: Any IPV Exposure and Adjusted Tower Density

5 Results

5.1 Baseline results

We present the baseline results in Table 4. Given the binary nature of the dependent variables, we report results from a Probit specification in columns (1)–(4). Column (1) presents estimates from a naïve specification where we regress the IPV outcomes on women's internet exposure without including any controls. Panels A to D correspond to different forms of IPV, viz. physical, sexual, emotional, and any IPV, respectively. Across all types of IPV, we observe a negative association between women's internet exposure and their probability of experiencing violence. As we move to columns (2) and (3), we sequentially add controls for women and household-level characteristics, and then for geographic and socio-demographic characteristics. Column (4) includes the full set of controls along with state fixed effects.[8] In this specification, internet exposure is associated with a statistically significant 4 percentage point (pp) reduction in the likelihood of physical IPV, 2 pp for emotional IPV, and 4 pp for experiencing any form of IPV. Columns (5)–(8) present the marginal effects from our instrumental variable (IV) estimation using Conditional Mixed Process (CMP), following the same progression of controls as in columns (1)–(4). Column (6) includes women and household characteristics, followed by district-level geographic and socioeconomic controls in column (7), and finally state fixed effects in column (8). Across all specifications, we find a statistically significant negative effect of internet exposure on IPV. Our preferred specification in column (8) includes the most comprehensive set of controls. In this model, internet exposure reduces the likelihood of physical violence by 17 pp (Panel A), sexual violence by 5 pp (Panel B), emotional violence by 9 pp (Panel C), and any form of IPV by 18 pp (Panel D). All effects are statistically significant at the 1% level, highlighting a strong causal relationship between digital connectivity and women's risk of domestic violence. These findings underscore the potential of digital access as a powerful tool in challenging entrenched norms and improving women's safety within their homes. By reducing women's exposure to IPV across multiple forms, internet access appears to disrupt cycles of control and isolation that often underpin domestic abuse. This is evidence, that in patriarchal contexts such as India, internet access can serve as a critical source of information, connection, and empowerment.

[8] When we include state fixed effects in column (4) and column (8) of Panel B, the sample size declines because STATA automatically drops observations that perfectly predict the outcome. It includes those observations in a state (Lakshadweep) in which either all women report experiencing sexual IPV or none do.

6 Potential mechanism

We next explore the mechanism that may explain the negative effect of women's internet exposure on their risk of experiencing IPV. We argue that changes in women's attitudinal beliefs about the acceptability of IPV constitute a channel of this effect. Specifically, we posit that access to information and awareness through internet connectivity can challenge and reshape internalized patriarchal norms. Women who are less likely to justify or accept IPV may be more empowered to resist violence or less likely to tolerate abusive behaviours, thereby reducing their actual exposure. Existing work suggests that digital connectivity can transform attitudes surrounding gender norms and domestic violence. As awareness increases, it is possible that such normative shifts within households foster an environment where violence is no longer accepted or normalized. In line with this, we focus on 'Any IPV' as our outcome, which captures a woman's experience of any form of physical, sexual, or emotional violence.

There is extensive evidence linking internet access to improvements in women's awareness, decision-making, and self-efficacy across domains such as healthcare, entrepreneurship, and self-esteem (Pandey et al., 2003; Meherali et al., 2021; Pickard et al., 2024; Thakkar et al., 2025). Building on this, we formally examine whether women's attitudes justifying IPV mediate the relationship between internet exposure and IPV outcomes. Following a two-step procedure similar to Mookerjee et al. (2023); Bose et al. (2020), we first estimate the effect of internet exposure on IPV attitudes. The NFHS-5 asks whether women believe a husband is justified in beating his wife under seven scenarios, *(1) if she goes out without telling him, (2) neglects the children, (3) argues with him, (4) refuses sex, (5) doesn't cook properly, (6) is unfaithful, and (7) is disrespectful.* We code each response as 1 if the woman justifies violence and 0 otherwise. Consistent with prior studies (Shreemoyee et al., 2025b; Dhamija et al., 2024), we construct a binary indicator, $IPV\,Attitudes$, equal to 1 if the woman justifies wife-beating in any of these situations, and 0 otherwise. We then estimate the effect of internet exposure on $IPV\,Attitudes$ using an IV specification, instrumenting internet exposure with the adjusted tower density at the district level. Table 5, column (1) shows that internet exposure significantly reduces the likelihood of justifying IPV by 21 pp.

In the second step, we estimate the association between women's IPV attitudes and their IPV exposure. We use the predicted values of $IPV\,Attitudes$ from the first step in a linear regression to examine whether a reduction in justifying attitudes is associated with lower actual IPV exposure. To address the generated regressor problem, we bootstrap the standard errors with 100 replications. The results reveal a positive and statistically significant association between IPV attitudes and IPV exposure. That is, women who are less likely to justify IPV are also less likely to experience it. Specifically, we find that a reduction in IPV-justifying attitudes is significantly associated with a lower likelihood of exposure to IPV, suggesting that attitudes around regressive gender norms indeed serve as a mechanism.

7 Heterogeneity analysis

While our average effects indicate a strong protective role of internet access against IPV, these impacts may not be uniform across all groups. Focusing solely on average treatment effects can mask meaningful heterogeneity based on individual, household, and regional characteristics. We therefore explore sub-sample heterogeneity by women's education, household wealth, social group, area of residence, and by regional classifications of Indian states. For each sub-sample, we re-estimate the effect of internet exposure on the 'Any IPV' outcome, including the full set of covariates and state fixed effects from our preferred specification. We summarize the results in Table 6, with Panel A focusing on individual and household-level characteristics and Panel B showing results across regional classifications.

7.1 By Individual and Household Characteristics

From Panel A of Table 6, we find significant heterogeneity in the effect of internet access on IPV exposure. We note that the negative effects of internet exposure on IPV are statistically significant across both backward and other social groups, and across rural and urban areas, all at the 1% level. The magnitudes are notably larger for women from backward castes and for those residing in urban areas. Among wealthier households, we find that internet exposure is leads to a significant 18 pp reduction in IPV exposure, significant at the 1% level. Among poorer households, we also observe a negative coefficient, but the effect is statistically significant only at the 10% level, and the standard errors are nearly double those for non-poorer households. This suggests that the lack of precision could stem from inflated standard errors, rather than the absence of an effect. Similarly, we find a 21 pp reduction in IPV exposure among more educated women, significant at 1% level. In contrast, the effect is not precisely estimated for less educated women, again accompanied by larger standard errors.[9]

7.2 By regional classification of states

India's socio-cultural diversity is reflected in substantial regional variation in gender norms, autonomy, and violence (Jejeebhoy and Sathar, 2001). Therefore, we examine heterogeneity by

[9]The limited statistical significance across some margins groups may reflect lower statistical power rather than a true absence of effect.

grouping states into standard regional classifications, viz., North, Central, East, West, South, North-East, and Empowered Action Group (EAG) states. We also study heterogeneity in the results by excluding the North-Eastern states and Union Territories from the sample. We present the results in Panel B of Table 6. Our findings reveal a strong and statistically significant effects in the South, Central, and Eastern states, as well as across EAG states, and the sample excluding NE states and UTs. In contrast, we detect no significant effects in the NE region, and only a weakly significant effect at the 10% level in Northern India. The absence or attenuation of effects in these regions may reflect baseline differences in women's mobility, digital access, or empowerment levels. For instance, women in parts of the North-East and Northern India may already enjoy relatively higher autonomy, reducing the marginal impact of internet exposure on IPV risk.

Our heterogeneity analysis highlights that the protective effects of internet exposure on IPV are heterogeneous across different margins. They appear to more precisely estimated among relatively privileged sub-groups, such as wealthier and educated and possibly more socially mobile women, and in certain geographic regions. However, in sub-samples where effects are not statistically significant, standard errors are considerably larger. This suggests limited precision, rather than a true null effect. From a policy perspective, this perhaps suggests that internet access alone is not sufficient to reduce women's likelihood of facing IPV and complementary strategies to expanding targeted digital access initiatives that reach less privileged women may be required.

8 Robustness

8.1 Trimmed instrumental variable

We test the robustness of our baseline results by trimming our instrumental variable, mobile tower density at district level, by top and bottom 1%, 5% and 10% of the IV respectively. We do this to test whether our baseline average treatment effects are driven by extreme values in cell tower density at district level. We present these results in Table 7, columns (1), (2) and (3), respectively. Our results remain robust across all three trimmed specifications. Column (1) records the effects of women's exposure to their risk of exposure to any IPV declines by 17 pp. Column (2) shows that women's exposure to IPV declines by 14 pp and column (3) reveals a decline in women's risk of being exposed to IPV by 16 pp if the woman has internet exposure. These results present evidence that our baseline effects are not driven by outliers in tower density across districts.

8.2 Additional controls

8.2.1 Husband and parental characteristics

Our baseline regressions account for a comprehensive set of controls, including individual-level characteristics of women, household characteristics, district-level geographic factors, and socioeconomic indicators, along with state fixed effects. To further assess the robustness of our results, we augment our baseline model by including husband and parental characteristics that may plausibly influence both a woman's access to the internet and her exposure to IPV. Specifically, we control for the husband's education, employment status, and frequency of alcohol consumption as these factors have been shown to correlate with household bargaining dynamics, IPV risk, and gender norms within the household (Field et al., 2004; Yount and Li, 2009; Krishnan et al., 2010; Shreemoyee et al., 2025b; Mookerjee et al., 2025). In addition, we include a covariate capturing whether the woman witnessed inter-parental violence during her formative years, an important predictor of IPV normalization and intergenerational transmission of violent norms (Mookerjee et al., 2021; Dhamija et al., 2024). We present the results in column (1) of Table 8. We find a robust and statistically significant effect. Specifically, we note a decline in women's risk of being exposed to any IPV by 15 pp if she has internet exposure, even after controlling for these additional controls.

8.2.2 Neighbourhood infrastructure

Here, we include covariates that capture contextual variation stemming from the neighbouring districts' infrastructure of cell tower (BTS) installations. Given that it is commonplace for women in rural India to relocate to nearby villages or districts after marriage, we worry about a potential violation of our exclusion restriction if women's reported internet exposure reflects the infrastructure of a different district than the one we use to construct our IV. For instance, if a woman reports her internet use prior to marriage and marriage migration, then the tower density in her current district no longer accurately predicts the woman's internet. Moreover, towers in neighbouring districts are likely to be correlated with the IV constructed from the tower density in her district of interview. To alleviate this concern, we also include as additional controls, the average tower density of neighbouring districts in our baseline specification and test the robustness of our results. Table 8, column (2) presents the results and we find a consistently similar story as in our baseline. We find that the causal effect of women's access to internet leads to an 18 pp decline in the probability of her being exposed to any IPV, even after the inclusion of this covariate, thereby reinforcing the credibility of our identification strategy.

8.2.3 Urbanisation redefined

In column (3) of Table 8, we alter the measurement of our control variable for urbanisation. In our baseline specification, one of the district-level socioeconomic controls is night-time lights intensity, which serves as a proxy for the level of urbanisation. As an alternative measure, we replace nightlights with the district-level Multidimensional Poverty Index (MPI), as reported in the National Multidimensional Poverty Index Report (NITI Aayog, 2023), to capture a broader conception of development and deprivation at the district level. We continue to find a consistently strong, robust, and negative effect of women's exposure to internet on her probability of being exposed to any kind of IPV (19 pp). The effect is statistically significant at 1% level of significance.

8.3 Alternative measures

Furthermore, our baseline outcomes consider physical, sexual, emotional IPV as well as a comprehensive measure that captures whether or not a woman is exposed to any of these forms of violence. Each of these outcomes, viz. physical, sexual and emotional IPV are based on several specific dimensions captured by the NFHS data.[10] While informative, our baseline binary indicators are not able to speak to the intensity of said forms of violence. Emotional violence largely based on verbal abuse is very different in its nature from intimate sexual or visible physical violence. Thus, we construct alternative outcomes using an Inverse-Probability-Weighted (IPW) index of the disaggregated components for physical, sexual, emotional and any IPV respectively. This approach weights the disaggregated components based on their prevalence in the sample, allowing us to construct a continuous intensity measures for each form of violence. We then create binary indicators of this intensity variable using the sample mean as the threshold. Thus, the binary variables now captures whether a woman's exposure to physical, sexual, emotional and any IPV lies above or below the average in the sample.

Table 9 presents the results. Column (1) shows that our baseline coefficients remain stable in their magnitude and significance. Women appear to continue to be at 17 (5 and 9) pp lower risk of physical (sexual and emotional) violence and 18 pp lower risk of any IPV, respectively. This reinforces the robustness of our causal findings.

8.4 Falsification test

Here, we address any concerns of spurious correlation in our findings and provide evidence to generate further confidence in our identification strategy. Our baseline results as well as

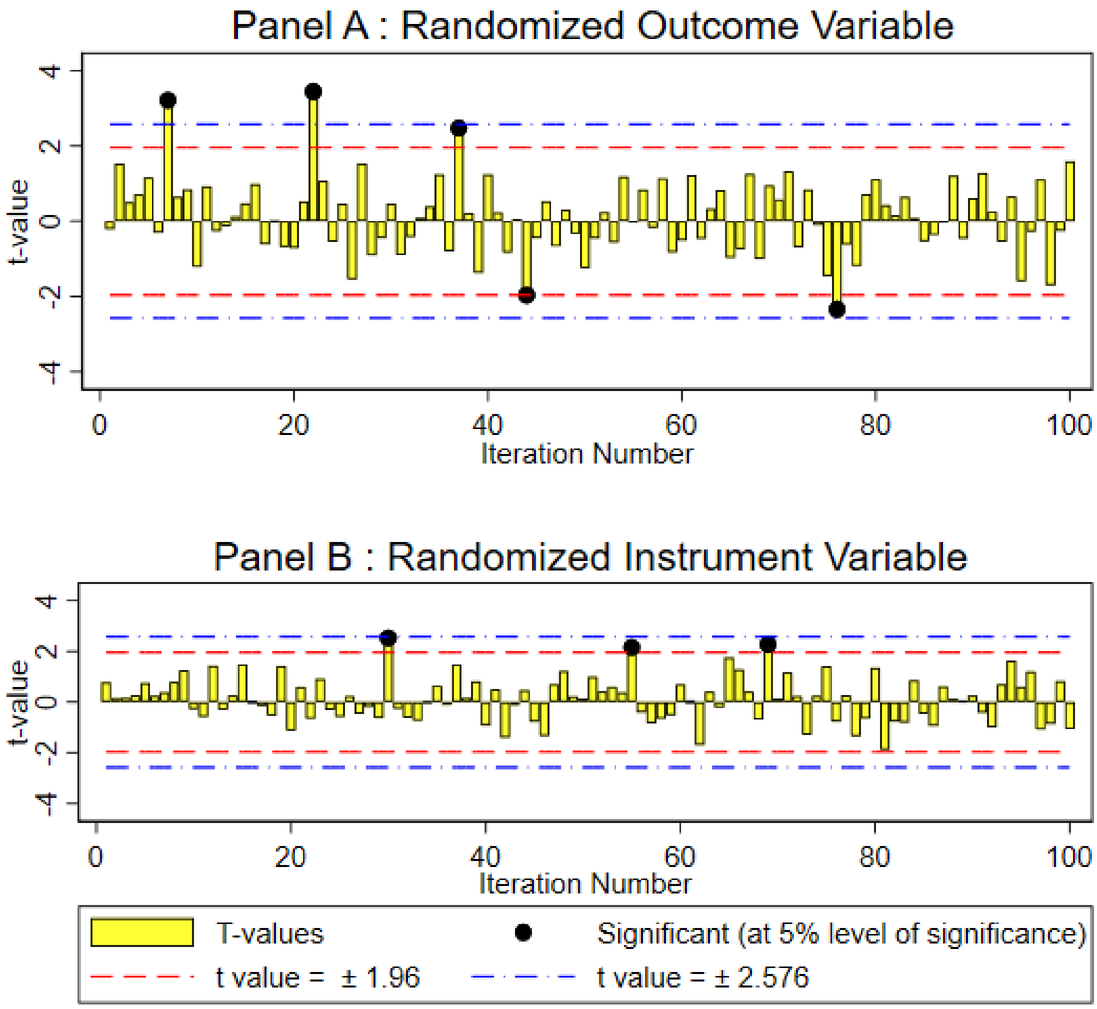
[10] See Table A1 for details on the specific acts of violence included in each category.

robustness results show consistently positive impact of internet exposure to women's position when it comes to IPV. This is captured in our negative and significant coefficients across several specifications. To test the credibility of these results, we conduct two tests of randomization. The first isolates the variation in our outcome variable and the second challenges the validity of our IV.

In the first falsification test, we retain the first-stage relationship between the instrument (district-level adjusted tower density) and the endogenous regressor (women's internet exposure), but randomly reassign the outcome variable (exposure to any IPV). That is, we associate a woman i's instrumented internet access with a randomly selected woman j's IPV exposure. In the second test, we randomize the instrumental variable by shuffling the adjusted tower density across districts. This disrupts the first-stage relationship by breaking the link between a woman's internet access and the actual tower density in her district.

We replicate both tests 100 times each and present the distribution of t-statistics in Panels A and B of Figure 5. In Panel A, we find that only 5 out of 100 times, we are unable to reject the null that the effect of internet exposure is equal to zero. This implies that 95% of the times, the effect of internet access on IPV exposure is not statistically significant when we consider another random woman's exposure to IPV. Panel B also shows that 97% of the times we do not find a statistically significant relationship between adjusted tower density and women's internet exposure when the IV is randomized. These results strongly suggest that our main estimates are not driven by random chance and support the validity of our causal identification strategy.

Figure 5: Falsification analysis



9 Conclusion

In this paper, we offer causal evidence on the empowering role of internet access in reducing women's exposure to intimate partner violence (IPV). Much has been written about how digital connectivity can transform women's lives, by expanding access to information, widening peer networks, and enhancing economic opportunities. We contribute to this growing literature by highlighting a less explored yet deeply consequential dimension, its potential to reduce Violence Against Women (VAW). Our work emphasizes a stark reality often documented in popular media and policy reports, that for many women, home is not a safe space. The threat of IPV not only violates women's basic human rights but also restricts their ability to act as independent decision-makers, pursue opportunities, and live with dignity. In such contexts, we posit that women's exposure to IPV should not be treated as an isolated outcome but as a critical marker of their broader autonomy, especially within patriarchal contexts.

Exploiting the rapid expansion of mobile tower infrastructure across India and using an Instrumental Variable strategy with district-level tower density as our instrument, we find that internet access significantly reduces the likelihood of women experiencing physical, emotional, and sexual violence by 17, 9, and 5 percentage points (pp) respectively. The likelihood of experiencing any kind of IPV reduces by 18 pp, from 25% to 7%. These findings are consistent across a wide range of additional controls, robustness checks, and falsification tests. According to Government of India projections (Ministry of Statistics and Programme Implementation, Government of India, 2022), the female population in 2021 was approximately 662 million. Assuming that roughly half of all women were married, our estimates suggest that greater exposure to the internet could have reduced the number of women experiencing any form of IPV from about 82.8 million to 23.2 million.

We also explore heterogeneity in our impacts across social and economic margins. The impact of internet access on IPV exposure is largely negative and statistically significant across caste, residence, wealth groups and women's education levels. However, the effects are particularly important among higher educated women and those from richer households. That said, the insignificant results for the less educated and marginally significant for poorer groups of women are perhaps indicative of lower precision in estimation and likely not reflective of a null effect given the large magnitudes. As such, to harness the full potential of digital access in addressing IPV, policy efforts ought to go beyond infrastructure expansion and invest in complementary digital policies. These policies could include strategies that boost digital literacy by providing more targeted awareness and contextual contents. Such efforts could then yield substantial dividends in improving women's safety and autonomy. Significant regional heterogeneity also exists in our impacts. While the impact is robust even in traditionally patriarchal regions like the EAG states, it is not statistically significant in the North-East of India perhaps reflecting regionally different gender norms where women may already enjoy higher social standing.

In conclusion, these findings provide evidence about the transformative potential of digital access for women's empowerment. From a policy standpoint, our findings suggest that efforts to bridge the digital divide should be seen not only as a matter of economic inclusion but also of safety and social justice. Promoting women's access to the internet can be a catalyst for change in the broader strategy to combat VAW. While digital connectivity is not a panacea, its ability to inform, connect, and empower women in ways that challenge deeply entrenched norms ought not to be underestimated.

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Table 1: Descriptive Statistics: Woman- and household-level variables

Variable	Mean	SD	Min	Max
Outcome Variables				
Physical IPV				
Yes	0.22	0.41	0.00	1.00
No	0.78	0.41	0.00	1.00
Sexual IPV				
Yes	0.05	0.21	0.00	1.00
No	0.95	0.21	0.00	1.00
Emotional IPV				
Yes	0.11	0.31	0.00	1.00
No	0.89	0.31	0.00	1.00
Any IPV				
Yes	0.25	0.43	0.00	1.00
No	0.75	0.43	0.00	1.00
Treatment Variable				
Internet use				
Yes	0.27	0.44	0.00	1.00
No	0.73	0.44	0.00	1.00
Women's Characteristics				
Age (in years)	34.09	8.00	15.00	49.00
Years of education	6.38	5.19	0.00	20.00
Currently married				
Yes	0.95	0.22	0.00	1.00
No	0.05	0.22	0.00	1.00
Employed during the last 12 months				
Yes	0.37	0.48	0.00	1.00
No	0.63	0.48	0.00	1.00
Media exposure				
Yes	0.73	0.44	0.00	1.00
No	0.27	0.44	0.00	1.00
Land/House ownership				
Yes	0.52	0.50	0.00	1.00
No	0.48	0.50	0.00	1.00
Duration of residence (in years)	14.88	10.26	0.00	49.00
Household Characteristics				
Male-headed household				
Yes	0.84	0.37	0.00	1.00
No	0.16	0.37	0.00	1.00
Age of household head (in years)	46.17	13.29	14.00	95.00
Number of household members	4.85	1.48	1.00	26.00
Religion				
Hindu	0.76	0.43	0.00	1.00
Muslim	0.12	0.32	0.00	1.00
Other	0.12	0.33	0.00	1.00
Social Group				
Scheduled Caste	0.19	0.39	0.00	1.00
Scheduled Tribe	0.19	0.40	0.00	1.00
OBC	0.38	0.49	0.00	1.00
Other/None	0.18	0.38	0.00	1.00
Don't Know/Missing	0.05	0.23	0.00	1.00
Wealth Category				
Poor	0.45	0.50	0.00	1.00
Non-poor	0.55	0.50	0.00	1.00
Residence				
Rural	0.76	0.43	0.00	1.00
Urban	0.24	0.43	0.00	1.00
Observations			61,262	

Notes: Women are considered exposed to mass media if they report exposure to any of the following: newspaper, radio, or television. Ownership of a house or agricultural land refers to either sole or joint ownership. Women who are widowed, divorced, separated, or not living with their husband are coded as currently unmarried. Other religions include Christian, Sikh, Buddhist/Neo-Buddhist, Jain, Jewish, Parsi/Zoroastrian, no religion, and others. The bottom two quantiles of the DHS wealth index (poorest and poor) are coded as poor, while the top three quantiles (middle, richer, and richest) are coded as non-poor.

Table 2: Descriptive Statistics: District-level variables

Variable	Mean	SD	Min	Max
Socioeconomic Characteristics				
Primary schools	1628.62	1125.47	17.00	7097.00
Middle schools	742.75	549.05	0.00	4133.00
Secondary schools	326.79	301.85	1.00	2762.00
Medical colleges	2.29	4.75	0.00	51.00
Engineering colleges	6.36	12.74	0.00	148.00
Arts and science colleges	14.75	17.17	0.00	113.00
Vocational schools	14.23	24.10	0.00	269.00
Family welfare centres	107.13	1058.19	0.00	7196.00
Maternity and child welfare centres	63.87	105.03	0.00	1799.00
Sex ratio (female/male)	0.95	0.06	0.53	1.18
SC and ST population share (in proportion)	0.33	0.23	0.00	0.99
Geographical Characteristics				
Ruggedness	1.94	0.76	0.96	4.05
Nightlights	0.60	0.53	0.00	3.88
Elevation	5.49	1.21	1.41	8.53
Number of lightning days	2.10	1.34	0.00	6.23
Vegetation Continuous Fields	17.82	15.22	0.00	71.81
Instrument Variable				
Number of towers	3123.51	7703.83	0.00	126593.00
Tower density	3.07	18.93	0.00	249.69
Adjusted tower density	0.49	0.77	0.00	5.52
Observations		692		

Notes: To minimise the influence of extreme values, the following variables ruggedness nightights, elevation, and number of lightning days, are expressed in their natural logarithmic form after adding one to them. Similarly, adjusted tower density is computed as log(1+tower density). Tower density represents the number of towers within a district divided by its land area (in per squares km).

Table 3: First stage analysis: Estimates of the relationship between tower density and women's internet exposure.

	Internet Exposure			
	(1)	(2)	(3)	(4)
Adjusted Tower Density	0.11***	0.03***	0.05***	0.08***
	(0.01)	(0.01)	(0.01)	(0.01)
Observations	61,262	61,262	61,262	61,262
Women's characteristics	No	Yes	Yes	Yes
Household characteristics	No	Yes	Yes	Yes
Geographical characteristics	No	No	Yes	Yes
Socioeconomic Characteristics	No	No	Yes	Yes
State fixed effects	No	No	No	Yes
Kleibergen–Paap rk LM statistic	161.43	37.44	22.54	41.37
Cragg–Donald Wald F statistic	3575.80	389.96	90.63	152.99
Kleibergen–Paap Wald rk F statistic	291.19	44.09	23.16	42.97
R-squared	0.06	0.33	0.34	0.35

Notes: Estimation via OLS regression analysis. Column (1) has no additional controls. Column (2) includes women- and household-level characteristics. Column (3) additionally adjusts for geographical and socioeconomic characteristics, while Column (4) further includes state fixed effects. Women's characteristics include age, current marital status, years of completed education, duration of residence at the current location, employment status (last 12 months), media exposure, and land/house ownership. Household characteristics include indicators for the sex of the household head, age of the household head, household size, religion, social group, wealth status, and area of residence. District-level socioeconomic characteristics include the sex ratio, proportion of SC and ST population, number of primary schools, middle schools, secondary schools, medical colleges, engineering colleges, arts and science colleges, vocational colleges, family welfare centres, maternity and child welfare centres. Geographical controls include district-level ruggedness, nightlights, elevation, annual number of days with lightning, and vegetation continuous fields. Standard errors (in parentheses) are clustered at the neighbourhood level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 4: Probit and IV analysis: Estimates of the effect of women's internet exposure on intimate partner violence.

	Probit				IV			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
				Panel A: Physical IPV				
Average Marginal Effect	-0.12***	-0.05***	-0.05***	-0.04***	-0.24***	-0.16***	-0.17***	-0.17***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.05)	(0.03)	(0.03)	(0.03)
Observations	61,262	61,262	61,262	61,262	61,262	61,262	61,262	61,262
				Panel B: Sexual IPV				
Average Marginal Effect	-0.03***	-0.01	-0.01	-0.00	-0.09***	-0.07***	-0.06***	-0.05***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.02)	(0.02)	(0.02)	(0.02)
Observations	61,262	61,262	61,262	61,174	61,262	61,262	61,262	61,174
				Panel C: Emotional IPV				
Average Marginal Effect	-0.05***	-0.02***	-0.02***	-0.02***	-0.06*	-0.07***	-0.08***	-0.09***
	(0.00)	(0.01)	(0.01)	(0.01)	(0.03)	(0.02)	(0.02)	(0.02)
Observations	61,262	61,262	61,262	61,262	61,262	61,262	61,262	61,262
				Panel D: Any IPV				
Average Marginal Effect	-0.12***	-0.05***	-0.05***	-0.04***	-0.24***	-0.17***	-0.19***	-0.18***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.05)	(0.03)	(0.03)	(0.03)
Observations	61,262	61,262	61,262	61,262	61,262	61,262	61,262	61,262
Women's characteristics	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Household characteristics	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Geographical characteristics	No	No	Yes	Yes	No	No	Yes	Yes
Socioeconomic Characteristics	No	No	Yes	Yes	No	No	Yes	Yes
State fixed effects	No	No	No	Yes	No	No	No	Yes

Notes: Columns (1) - (4) presents the findings from the Probit estimation, whereas columns (5) - (8) reports the results from the Conditional Mixed Process (CMP) estimation. Columns (1) and (5) has no additional controls. Columns (2) and (6) include women- and household-level characteristics. Columns (3) and (7) additionally adjust for geographical and socioeconomic characteristics, while columns (4) and (8) further include state fixed effects. Women's characteristics include age, current marital status, years of completed education, duration of residence at the current location, employment status (last 12 months), media exposure, and land/house ownership. Household characteristics include indicators for the sex of the household head, age of the household head, household size, religion, social group, wealth status, and area of residence. District-level socioeconomic characteristics include the sex ratio, proportion of SC and ST population, number of primary schools, middle schools, secondary schools, medical colleges, engineering colleges, arts and science colleges, vocational colleges, family welfare centres, maternity and child welfare centres. Geographical controls include district-level ruggedness, nightlights, elevation, annual number of days with lightning, and vegetation continuous fields. Standard errors (in parentheses) are clustered at the cluster level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 5: Mechanism: Estimates of the effect of women's internet exposure on their attitudes towards IPV.

	IV	OLS
	Attitudes towards IPV	Any IPV
	(1)	(2)
Internet Use	-0.21^{***}	
	(0.03)	
Fitted attitudes towards IPV		0.08***
		(0.02)
Observations	60,841	60,841
Women's Characteristics	Yes	Yes
Household Characteristics	Yes	Yes
Geographical Characteristics	Yes	Yes
Socioeconomic Characteristics	Yes	Yes
State Fixed Effects	Yes	Yes

Notes: Estimation in column (1) and (2) via CMP and OLS regression analysis, respectively. Both the columns include include women- and household-level characteristics, geographical characteristics, socioeconomic characteristics, and state fixed effects. Women's characteristics include age, current marital status, years of completed education, duration of residence at the current location, employment status (last 12 months), media exposure, and land/house ownership. Household characteristics include indicators for the sex of the household head, age of the household head, household size, religion, social group, wealth status, and area of residence. District-level socioeconomic characteristics include the sex ratio, proportion of SC and ST population, number of primary schools, middle schools, secondary schools, medical colleges, engineering colleges, arts and science colleges, vocational colleges, family welfare centres, maternity and child welfare centres. Geographical controls include district-level ruggedness, nightlights, elevation, annual number of days with lightning, and vegetation continuous fields. Standard errors (in parentheses) are clustered at the cluster level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 6: Heterogeneity analysis based on women's characteristics, household characteristics and regions.

	Panel A: Women's and Household Characteristics							
	Education		Social Group		Wealth		Residence	
	Low	High	Backward	Others	Poor	Non-Poor	Rural	Urban
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Average Marginal Effect	-0.15	-0.21***	-0.20***	-0.12***	-0.13*	-0.18***	-0.18***	-0.22***
	(0.11)	(0.06)	(0.03)	(0.05)	(0.08)	(0.03)	(0.03)	(0.05)
Observations	28,794	32,454	47,088	10,840	27,645	33,531	46,570	14,625
	Panel B: Regions							
Panel B	North	Central	East	West	South	North-East	Empowered Action Group (EAG)	All States except North-East & UTs
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Average Marginal Effect	-0.08*	-0.19***	-0.14**	-0.11	-0.31***	0.10	-0.20***	-0.19***
	(0.04)	(0.06)	(0.07)	(0.09)	(0.05)	(0.08)	(0.04)	(0.03)
Observations	11,473	13,182	10,614	6,183	10,222	9,423	26,438	48,299
Women's Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographical Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Socioeconomic Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Estimation via Conditional Mixed Process (CMP) estimation. The outcome variable is women's exposure to Any IPV. Panel A reports subsample estimates by women's and household characteristics. Columns (1)–(2) present results for women with below-average (¡6.38 years) and above-average (6.38 years) education, respectively. Column (3) includes households belonging to SC, ST, or OBC, while Column (4) covers other social groups. Columns (5) and (6) report estimates for poor and non-poor households, respectively, based on the DHS wealth index. Columns (7) and (8) present results for rural and urban residents, respectively. Panel B reports subsample analyses by regional classifications. Column (1) corresponds to Northern states and Union Territories (UTs): Jammu and Kashmir, Himachal Pradesh, Punjab, Chandigarh, Uttarakhand, Haryana, Delhi, and Rajasthan. Column (2) includes the Central region (Uttar Pradesh, Chhattisgarh, Madhya Pradesh), Column (3) covers the Eastern region (Bihar, West Bengal, Jharkhand, Odisha), Column (4) the Western region (Gujarat, Dadra and Nagar Haveli and Daman and Diu, Maharashtra, Goa), Column (5) the Southern region (Andhra Pradesh, Telangana, Karnataka, Lakshadweep, Kerala, Tamil Nadu, Puducherry, Andaman & Nicobar Islands), Column (6) the Northeastern region (Sikkim, Arunachal Pradesh, Nagaland, Manipur, Mizoram, Tripura, Meghalaya, Assam), Column (7) the Empowered Action Group (EAG) states (Bihar, Jharkhand, Odisha, Chhattisgarh, Madhya Pradesh, Uttar Pradesh, Uttarakhand, Rajasthan), and Column (8) excludes Northeastern states and UTs. All specifications include women's characteristics, household characteristics, district-level socioeconomic and geographic controls, and state fixed effects. Standard errors (in parentheses) are clustered at the cluster level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 7: Robustness analysis by trimming the extreme values of the instrument variable.

	Any IPV		
	(1)	(2)	(3)
Average Marginal Effect	-0.17***	-0.14***	-0.16***
	(0.03)	(0.03)	(0.03)
Observations	59,799	55,088	48,931
Women's Characteristics	Yes	Yes	Yes
Household Characteristics	Yes	Yes	Yes
Geographical Characteristics	Yes	Yes	Yes
Socioeconomic Characteristics	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes

Notes: Estimation via Conditional Mixed Process (CMP) estimation. The outcome variable is women's exposure to Any IPV. Columns (1), (2), and (3) excludes the top as well as bottom 1%, 5%, and 10% of the adjusted tower density, respectively. All specifications include women's characteristics, household characteristics, district-level socioeconomic and geographic controls, and state fixed effects. Standard errors (in parentheses) are clustered at the cluster level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 8: Robustness analysis by including additional or alternative controls.

	Any IPV		
	(1)	(2)	(3)
Average Marginal Effects	-0.16***	-0.18***	-0.19***
	(0.03)	(0.03)	(0.03)
Observations	58,007	60,685	61,045
Women's Characteristics	Yes	Yes	Yes
Household Characteristics	Yes	Yes	Yes
Geographical Characteristics	Yes	Yes	Yes
Socioeconomic Characteristics	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes
Husband & Parental Characteristics	Yes	No	No
Neighbouring District IV Average	No	Yes	No
Multidimensional Poverty Index	No	No	Yes

Notes: Estimation via Conditional Mixed Process (CMP) estimation. The outcome variable is women's exposure to Any IPV. All specifications include women's characteristics, household characteristics, district-level socioeconomic and geographic controls, and state fixed effects. Additionally, column (1) includes husband and parental characteristics: husband's education, employment in the last 12 months, age, and frequency of drinking as well has woman's exposure to inter-parental violence (=1 if yes and 0 otherwise). Column (2) includes the average tower density of all the neighbouring districts as an additional control. In column (3), we have replaced the nightlights with an alternative definition of urbanisation measured by district level Multidimensional Poverty Index. Standard errors (in parentheses) are clustered at the cluster level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 9: Robustness analysis based on alternative outcome variables

	Physical IPV	Sexual IPV	Emotional IPV	Any IPV
	(1)	(2)	(3)	(4)
Average Marginal Effect	-0.17***	-0.05***	-0.09***	-0.18***
	(0.03)	(0.02)	(0.02)	(0.03)
Observations	61,262	61,174	61,262	61,262
Women's Characteristics	Yes	Yes	Yes	Yes
Household Characteristics	Yes	Yes	Yes	Yes
Geographical Characteristics	Yes	Yes	Yes	Yes
Socioeconomic Characteristics	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes

Notes: Estimation via Conditional Mixed Process (CMP) estimation. Each measure of IPV in columns (1) to (4) is reconstructed based on an Inverse-Probability-Weighted (IPW) index of the disaggregated components for physical, sexual, emotional and any IPV. This index is then converted into a binary measure using the sample mean as the cut-off of the cutoff. Thus, the reconstructed binary measure of IPV now captures whether a woman's exposure to physical, sexual, emotional and any IPV lies above or below the average value of the IPW index. All specifications include women's characteristics, household characteristics, district-level socioeconomic and geographic controls, and state fixed effects. Standard errors (in parentheses) are clustered at the cluster level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Appendix

Table A1: Violences underlying each IPV

Type of IPV	Underlying Violence
Physical IPV	Acts of pushing, shaking, throwing something, twisting arm, pulling hair, slapping, punching with fist or something else, kicking, beating, choking, burning, threatening or attacking with any kind of weapon by partner
Sexual IPV	Forced sexual acts or humiliating sexual acts forced by partner
Emotional IPV	Partner's activities leading to humiliation, insult, or various kinds of threats to hurt wife or her closed ones

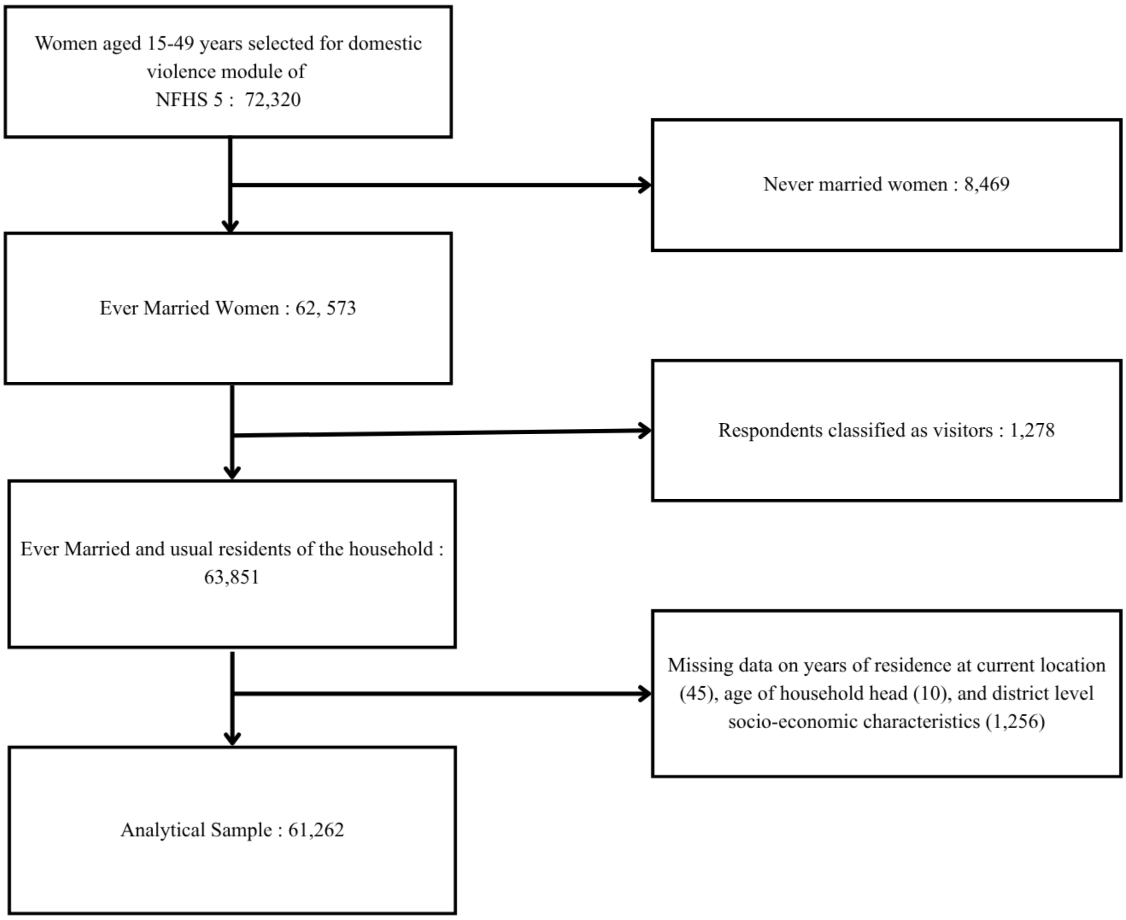

Figure A1: Analytical Sample formation